

MSCS DEPT COLLOQUIUM:

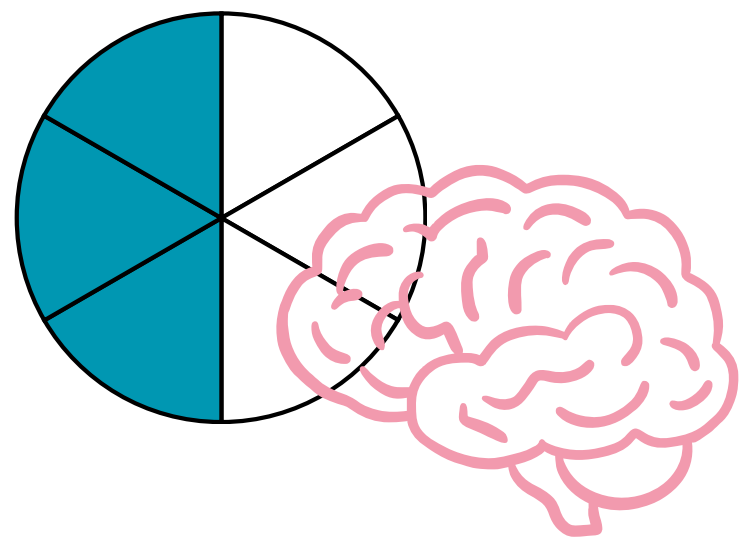
PARTS AND WHOLE: DECODING FRACTION PROCESSING INSIGHTS FROM EEG BRAIN SIGNALS

MONDAY, MARCH 23, 2026 | 3:30PM

SPEAKER BIO: Dr. Brian Rivera is a Visiting Assistant Professor of Psychology at St. Olaf College. He earned his PhD in Educational Psychology from the University of Alabama with a concentration in Educational Neuroscience. His research uses EEG to investigate how the brain processes numerical information, with a particular focus on fraction processing and multivariate pattern analysis. Previously, he worked on the Baby's First Years study at the University of Nebraska Lincoln, which examined how poverty affects brain development in early childhood.

Our Speaker:
Brian Rivera

VISITING ASSISTANT PROFESSOR
OF PSYCHOLOGY AT ST. OLAF



When you see the fraction $\frac{3}{6}$, does your brain immediately recognize it as one-half, or does it first process the 3 and the 6 separately? This question is at the heart of a longstanding debate in numerical cognition about whether fraction processing is holistic, componential, or both. This talk presents EEG research that uses machine learning (multivariate pattern analysis - MVPA) to decode how the brain represents fractions over time. Using a series of classification analyses on brain activity recorded during a fraction verification task, results show that both magnitude and component information are decodable from the EEG signal, but with distinct time courses, supporting a hybrid model of fraction processing. The talk will discuss what these findings mean for theories of numerical cognition and how MVPA can complement traditional approaches for studying math in the brain.

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